EcoRide Fleet Platform — MVP Technical Specification

Based on the Phase 1 MVP scope: **driver onboarding, agreement signing, shift scheduling, notifications, and earnings visibility**.

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1. MVP objective

Build a working first version of the EcoRide platform that allows:

* drivers to register and complete onboarding
* drivers to upload compliance documents
* drivers to sign the active agreement
* admins to review and approve drivers
* admins to create and manage shifts
* drivers to view and request shifts
* admins or system to confirm shift allocations
* admins to create/import and publish earnings
* drivers to view published earnings
* drivers to receive notifications and reminders

This MVP excludes advanced vehicle control, biometrics, telematics, AI allocation, and live GPS operations.

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2. System architecture split

■ Mobile App (Driver App)

Used by drivers for:

* account registration and login
* onboarding
* document upload
* agreement signing
* viewing application status
* viewing and requesting shifts
* viewing confirmed shifts
* viewing earnings
* receiving notifications
* managing their own profile

■■ Web App (Admin Dashboard)

Used by EcoRide operations/admin staff for:

* admin login
* reviewing driver applications
* approving / rejecting / suspending drivers
* reviewing documents
* managing agreements
* creating and editing shifts
* reviewing and managing shift requests/allocations
* creating/importing/publishing earnings
* sending notifications
* viewing operational summaries

Used by both apps:

* data storage / collections
* permissions
* validations
* business logic
* workflows
* reminder automations
* status transitions

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3. User roles

3.1 Driver

Can:

* register and log in
* complete onboarding
* upload documents
* sign agreements
* view own application status
* view open shifts
* request shifts
* view own confirmed shifts
* view own earnings
* receive notifications

Cannot:

* approve own account
* edit earnings
* manage shifts globally
* view other drivers’ data

3.2 Admin / Operations

Can:

* log in to admin dashboard
* review all drivers
* approve, reject, suspend drivers
* review documents and signatures
* manage agreements
* create/edit/close/cancel shifts
* manage allocations
* create/import/publish earnings
* send notifications
* view operational dashboards

These collections are shared across both the mobile and web apps.

4.1 Users

Purpose

Authentication and system role record.

Fields

* `id`
* `role` — enum: `driver`, `admin`
* `email` — string, unique, required
* `phone` — string, unique, required
* `password_hash` — auth managed
* `is_active` — boolean
* `status` — enum:

* `pending`
* `active`
* `suspended`
* `rejected`
* `last_login_at` — datetime
* `created_at` — datetime
* `updated_at` — datetime

4.2 DriverProfiles

Purpose

Stores all driver-specific profile and compliance information.

Fields

* `id`
* `user_id` — relation to Users
* `first_name`
* `last_name`
* `id_number`
* `date_of_birth`
* `gender`
* `phone_alt`
* `email_alt`
* `residential_address_line_1`
* `residential_address_line_2`
* `city`
* `province`
* `postal_code`
* `bank_name`
* `bank_account_holder`
* `bank_account_number`
* `bank_branch_code`
* `account_type` — enum:

* `cheque`
* `savings`
* `transmission`
* `bolt_driver_status_confirmed` — boolean
* `onboarding_status` — enum:

* `not_started`
* `in_progress`
* `submitted`
* `under_review`
* `approved`
* `rejected`
* `agreement_status` — enum:

* `not_signed`
* `signed`
* `documents_status` — enum:

* `missing`
* `partial`
* `submitted`

* `approved`
* `rejected`
* `application_submitted_at`
* `approved_at`
* `approved_by`
* `rejection_reason`
* `notes_internal`
* `created_at`
* `updated_at`

Derived / computed flags

* `is_profile_complete`
* `is_documents_complete`
* `is_eligible_for_shift_selection`

4.3 DriverDocuments

Purpose

Stores uploaded driver documents.

Fields

* `id`
* `driver_profile_id`
* `document_type` — enum:

* `drivers_license`
* `pdp`
* `profile_photo`
* `id_document`
* `proof_of_address`
* `bank_confirmation`
* `other`
* `file_url`
* `file_name`
* `file_size`
* `mime_type`
* `document_number`
* `expiry_date`
* `verification_status` — enum:

* `pending`
* `approved`

* `rejected`
* `review_notes`
* `uploaded_at`
* `reviewed_at`
* `reviewed_by`

MVP required documents

* driver’s license
* PDP
* ID document
* profile photo

4.4 Agreements

Purpose

Stores agreement versions/templates.

Fields

* `id`
* `title`
* `version`
* `content_html`
* `content_pdf_url`
* `is_active`
* `effective_date`
* `created_by`
* `created_at`
* `updated_at`

Rule

Only one active agreement version can be active at a time.

4.5 AgreementSignatures

Purpose

Stores a driver’s signed agreement record.

Fields

* `id`
* `agreement_id`
* `driver_profile_id`
* `agreed_full_name`
* `signature_data`
* `accepted_at`
* `ip_address`
* `device_info`
* `version_signed`
* `signed_record_pdf_url`
* `created_at`

Rule

Driver must have a valid signature for the active agreement before shift access is allowed.

4.6 Shifts

Purpose

Stores operational shifts.

Fields

* `id`
* `title`
* `shift_date`
* `start_time`
* `end_time`
* `location_name`
* `location_address`
* `capacity`
* `status` — enum:

* `draft`
* `open`
* `full`
* `closed`
* `cancelled`
* `notes`
* `created_by`

* `created_at`
* `updated_at`

Derived fields

* `allocated_count`
* `remaining_slots`

4.7 ShiftAllocations

Purpose

Stores driver requests and allocations.

Fields

* `id`
* `shift_id`
* `driver_profile_id`
* `status` — enum:

* `requested`
* `confirmed`
* `declined`
* `cancelled`
* `waitlisted`
* `requested_at`
* `confirmed_at`
* `declined_at`
* `cancelled_at`
* `decline_reason`
* `admin_notes`
* `created_by`

Rules

* driver cannot request same shift twice
* driver cannot exceed weekly shift limit
* only eligible drivers can request
* allocation cannot exceed capacity

4.8 Earnings

Purpose

Stores earnings by shift/driver.

Fields

* `id`
* `driver_profile_id`
* `shift_id`
* `earning_date`
* `gross_earnings`
* `deductions_total`
* `incentives_total`
* `penalties_total`
* `net_earnings`
* `trips_completed`
* `earnings_source` — enum:

* `manual_entry`
* `csv_import`
* `api_import`
* `source_reference`
* `breakdown_json`
* `status` — enum:

* `draft`
* `published`
* `created_by`
* `created_at`
* `updated_at`

4.9 Notifications

Purpose

Stores notifications and delivery state.

Fields

* `id`
* `user_id`
* `title`
* `message`
* `notification_type` — enum:

* `application`
* `shift`
* `earnings`
* `system`
* `reminder`
* `related_entity_type` — enum:

* `driver_profile`
* `shift`
* `earning`
* `agreement`
* `general`
* `related_entity_id`
* `delivery_channel` — enum:

* `in_app`
* `push`
* `sms`
* `delivery_status` — enum:

* `pending`
* `sent`
* `failed`
* `read`
* `scheduled_for`
* `sent_at`
* `read_at`
* `created_at`

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5. ■ Mobile app (Driver app) specification

This section defines everything the driver uses.

5.1 Mobile app navigation

Primary nav:

* Dashboard
* Shifts
* Earnings
* Notifications
* Profile

Onboarding pages may sit outside the main nav until complete.

5.2 Splash / Loading page

Purpose

Check authentication state and route appropriately.

UI elements

* logo
* loading spinner

Logic

* unauthenticated → Login
* authenticated + onboarding incomplete → Onboarding Home
* authenticated + pending/restricted → Account Status
* authenticated + approved → Driver Dashboard

5.3 Login page

Fields

* email or phone
* password

Buttons

* Log in
* Go to Register
* Forgot password

Validation

* required fields
* valid email format if email used

States

* loading
* invalid credentials
* suspended account
* rejected account

5.4 Register page

Fields

* first name
* last name
* email
* phone
* password
* confirm password
* checkbox: accept privacy/terms

Buttons

* Create account
* Back to login

Validation

* email unique
* phone unique
* passwords match
* minimum password strength
* privacy checkbox required

Action

On submit:

* create User
* create DriverProfile
* set onboarding_status = not_started
* send welcome notification
* redirect to onboarding

5.5 Forgot password page

Fields

* email or phone

Button

* Send reset link / code

5.6 Account Status page

Purpose

Show drivers who are pending, rejected, or suspended where they stand.

Display

* status badge
* explanation text
* rejection reason if applicable
* support contact block

Buttons

* Continue onboarding
* Contact support
* Log out

5.7 Onboarding Home page

Purpose

Central onboarding progress page.

Display

* progress indicator
* step checklist:

* personal details
* documents
* agreement
* submit application
* application status

Buttons

* Continue profile
* Upload documents
* Sign agreement
* Review and submit

Rules

Submit enabled only when all required sections are complete.

5.8 Personal Details page

Sections and fields

Identity

* first name
* last name
* ID number
* date of birth

Contact

* email
* phone
* alternate phone

Address

* address line 1
* address line 2
* city
* province
* postal code

Banking

* bank name
* account holder name
* account number
* account type
* branch code

Driver declaration

* checkbox: I am an active Bolt driver

Buttons

* Save draft
* Continue

Validation

* core identity required
* core address required
* banking required before submission
* Bolt driver declaration required before submission

5.9 Documents Upload page

Purpose

Upload required compliance documents.

Required upload blocks

* Driver’s license
* PDP
* ID document
* Profile photo

Optional upload blocks

* Proof of address
* Bank confirmation
* Other

Each upload block contains

* label
* upload button
* replace file action
* file name preview
* verification status badge
* review notes if rejected

Buttons

* Save draft
* Continue

Validation

Required documents must exist before application can be submitted.

5.10 Agreement Review page

Display

* agreement title
* version
* effective date
* full scrollable agreement content

Controls

* checkbox: I have read and understood this agreement
* Proceed to sign button

Validation

Checkbox required before moving forward

5.11 Agreement Signature page

Fields

* full legal name
* signature pad
* checkbox: I agree to the terms
* auto date/time display
* agreement version display

Actions

* clear signature
* submit signature
* back

Result

* create signature record
* mark agreement as signed
* return success state

5.12 Application Review & Submit page

Display

* personal details summary
* document checklist
* agreement signed confirmation
* application completeness summary

Buttons

* Edit personal details
* Edit documents
* Edit agreement
* Submit application

Validation

Can only submit if:

* profile complete
* required docs uploaded
* agreement signed

Action

On submit:

* set onboarding_status = submitted
* set application_submitted_at = now
* trigger admin notification
* show confirmation

5.13 Driver Dashboard page

Display cards

* welcome card

* account status
* next confirmed shift
* shifts this week
* weekly earnings summary
* latest notifications

Buttons

* View shifts
* My shifts
* Earnings
* Notifications
* Profile

Conditional logic

If driver not approved, show onboarding/pending state instead of standard dashboard functionality.

5.14 Available Shifts page

Filters

* week selector
* date
* location optional

Shift card content

* title
* date
* start/end times
* location
* remaining slots
* status badge

Actions

* Request shift

Validation

* driver must be active
* onboarding approved

* agreement signed
* weekly limit not exceeded
* no duplicate request
* shift open
* shift not in past
* capacity available

5.15 My Shifts page

Tabs

* Requested
* Confirmed
* Past

Per shift card

* title
* date
* start/end time
* location
* status
* notes

Actions

* cancel request if permitted
* view details

5.16 Shift Detail page

Display

* shift title
* date
* start/end
* location
* notes
* allocation/request status

Actions

* request shift
* cancel request
* back

5.17 Earnings List page

Filters

* week
* month
* date range

Per earning row/card

* date/shift
* trips completed
* gross
* deductions
* incentives
* net
* status

Action

* View details

Rule

Only published earnings are visible to drivers.

5.18 Earnings Detail page

Display

* date / shift
* trips completed
* gross
* deductions
* incentives
* penalties

* net earnings
* detailed breakdown
* notes

5.19 Notifications page

Display

* notification list
* title
* preview text
* timestamp
* read/unread state

Actions

* open notification
* mark as read

5.20 Profile page

Sections

* personal details
* contact details
* address
* banking details
* document statuses
* agreement status
* account status

Actions

* edit profile fields where allowed
* re-upload rejected document
* log out

Restriction

Sensitive changes after submission/approval may require admin review.

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6. ■■ Web app (Admin dashboard) specification

This section defines everything the operations/admin team uses.

6.1 Admin dashboard navigation

Primary nav:

* Dashboard
* Drivers
* Agreements
* Shifts
* Earnings
* Notifications
* Settings

6.2 Admin Login page

Fields

* email
* password

Button

* Log in

6.3 Dashboard Home page

KPI cards

* total drivers
* pending applications
* approved drivers
* open shifts this week
* shifts requiring allocation
* earnings pending publish

Tables/widgets

* recent applications
* upcoming shifts
* recent activity

6.4 Drivers List page

Filters

* onboarding status
* user status
* documents status
* agreement status
* search by name, phone, email, ID

Table columns

* full name
* phone
* email
* onboarding status
* documents status
* agreement status
* user status
* submitted at

Actions

* View profile
* Approve
* Reject
* Suspend

6.5 Driver Detail page

Sections

Header summary

* full name
* internal ID

* phone
* email
* status badges

Personal details

* full submitted profile

Banking details

* bank info

Documents section

* file previews/links
* status per doc
* approve/reject controls
* review notes

Agreement section

* version signed
* signed at
* signature preview
* signed record link

Shift history

* requested shifts
* confirmed shifts
* past shifts

Earnings summary

* recent earnings
* total earnings snapshot

Internal notes

* notes field

Actions

* Approve driver
* Reject driver
* Suspend driver
* Request more information
* Send notification

6.6 Driver Approval / Rejection modal

Approve action

Updates:

* Users.status = active
* DriverProfiles.onboarding_status = approved
* DriverProfiles.approved_at = now
* DriverProfiles.approved_by = current admin

Creates notification:

* application approved

Reject action

Field:

* rejection reason required

Updates:

* Users.status = rejected
* DriverProfiles.onboarding_status = rejected
* DriverProfiles.rejection_reason = entered reason

Creates notification:

* application rejected

6.7 Agreements Management page

Table columns

* title
* version
* effective date
* active
* created at

Actions

* Create agreement
* Edit agreement
* Activate/deactivate agreement

Create/Edit fields

* title
* version
* effective date
* content HTML/rich text
* upload PDF optional
* active toggle

Rule

Only one active agreement at a time.

6.8 Shifts List page

Filters

* status
* date range
* location

Table columns

* title
* shift date
* start time
* end time
* capacity
* allocated count
* remaining slots
* status

Actions

* Create shift
* Edit shift
* View allocations
* Close shift
* Cancel shift

6.9 Create / Edit Shift page

Fields

* title
* date
* start time
* end time
* location name
* location address
* capacity
* notes
* status

Validation

* date required
* start < end
* capacity minimum 1

6.10 Shift Detail page

Display

* shift summary
* capacity
* allocated count
* remaining slots

Tabs

* Requested
* Confirmed
* Waitlisted
* Cancelled

Per row

* driver name
* phone
* request status
* request time
* admin notes

Actions

* confirm request

* decline request
* move to waitlist
* cancel allocation
* send message

Rule

When confirmed count reaches capacity, shift auto-marks full.

6.11 Earnings List page

Filters

* date range
* driver
* shift
* status
* source

Table columns

* driver
* shift/date
* gross
* deductions
* incentives
* net
* trips
* status
* source

Actions

* Create earning
* Import CSV
* Edit earning
* Publish
* Unpublish

6.12 Create / Edit Earnings page

Fields

* driver
* shift
* earning date
* trips completed
* gross earnings
* deductions total
* incentives total
* penalties total
* net earnings
* source
* source reference
* breakdown
* status

Logic

If net not manually entered:
`net = gross - deductions - penalties + incentives`

Validation

* driver required
* date required
* gross required

6.13 CSV Import page

Fields / controls

* upload CSV
* map columns:

* driver identifier
* shift identifier/date
* gross
* deductions
* incentives
* penalties
* trips completed
* preview rows
* import button

Result

* success count
* failed rows
* error download/report

6.14 Notifications page

Functions

* view sent notifications
* create/send notifications

Send form fields

* recipient type:

* single driver
* multiple drivers
* all approved drivers
* title
* message
* delivery channel
* schedule send optional

Actions

* Send now
* Schedule

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7. Shared workflows / automations

7.1 Registration workflow

On register:

* create User
* create DriverProfile
* send welcome notification
* route user to onboarding

7.2 Profile completion workflow

On save:

* update DriverProfile
* recalculate `is_profile_complete`

7.3 Document upload workflow

On upload:

* create DriverDocument with verification_status = pending
* recalculate documents completeness

7.4 Agreement signing workflow

On submit signature:

* create AgreementSignature
* set DriverProfile.agreement_status = signed
* optionally create audit log

7.5 Application submit workflow

On submit application:

* validate profile complete
* validate required docs present
* validate agreement signed
* set onboarding_status = submitted
* notify admin team

7.6 Driver approval workflow

On approval:

* activate user
* set onboarding approved
* stamp approved_by and approved_at

* notify driver

7.7 Shift request workflow

On request:

* validate driver status active
* validate onboarding approved
* validate agreement signed
* validate weekly shift count
* validate shift open
* validate capacity
* validate shift not in past

Then either:

* create `requested` allocation for admin review
or
* auto-create `confirmed` allocation if using instant confirmation model

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7.8 Shift capacity workflow

* if confirmed allocations >= capacity → shift status = full
* if confirmed allocations < capacity and shift was full → shift status = open

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7.9 Earnings publish workflow

On publish:

* set earning status = published
* send driver notification that earnings are available

7.10 Shift reminder workflow

Time-triggered:

* 24 hours before confirmed shift
* 2 hours before confirmed shift

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8. Validation rules

Registration

* unique email
* unique phone
* password valid
* privacy acceptance required

Onboarding submission

* personal details complete
* address complete
* bank details complete
* active Bolt declaration complete
* required documents uploaded
* agreement signed

Shift request

* driver active
* onboarding approved
* agreement signed
* shift open
* shift in future
* no duplicate request
* weekly limit not exceeded

Earnings

* driver exists
* numeric values valid
* publish only when complete

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9. Status model

User status

* `pending`
* `active`
* `suspended`
* `rejected`

Onboarding status

* `not_started`
* `in_progress`
* `submitted`
* `under_review`
* `approved`
* `rejected`

Document verification status

* `pending`
* `approved`
* `rejected`

Agreement status

* `not_signed`
* `signed`

Shift status

* `draft`
* `open`
* `full`
* `closed`
* `cancelled`

Shift allocation status

* `requested`
* `confirmed`
* `declined`
* `cancelled`
* `waitlisted`

Earnings status

* `draft`
* `published`

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10. Permissions

Driver permissions

Can read:

* own profile
* own documents
* own agreement signature
* open shifts
* own allocations
* own earnings
* own notifications

Can write:

* own onboarding info
* own uploads
* own shift requests
* own signature
* own allowed profile edits

Admin permissions

Can read/write:

* all drivers
* all documents
* all agreements
* all shifts
* all allocations
* all earnings
* all notifications

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11. What belongs where

■ Mobile app only

* register/login
* onboarding pages
* documents upload
* agreement signing
* driver dashboard
* available shifts
* my shifts
* earnings pages
* notifications list
* profile page

■■ Web app only

* admin dashboard
* drivers list/detail
* driver approval/rejection/suspension
* agreements management
* shift creation/editing
* allocation controls
* earnings management
* CSV imports
* bulk notifications

■ Shared backend only

* collections
* workflows
* validation rules
* permissions
* calculations
* reminder automation
* status transitions

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12. Explicit MVP exclusions

Do not build in MVP:

* facial recognition
* keyless vehicle unlock
* telematics integration
* live location tracking
* incident escalation engine
* AI shift allocation
* driver scoring/tiering
* predictive analytics
* vehicle readiness/fuel/maintenance engine

These are later phases, not Phase 1 MVP.

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13. Completion criteria

The MVP is complete when this full loop works:

1. Driver registers in mobile app
2. Driver completes onboarding in mobile app
3. Driver uploads required documents in mobile app

4. Driver signs agreement in mobile app
5. Driver submits application in mobile app
6. Admin reviews and approves driver in web app
7. Admin creates shifts in web app
8. Driver views and requests shifts in mobile app
9. Admin or system confirms shifts via web app/backend
10. Admin creates/imports and publishes earnings in web app
11. Driver views earnings in mobile app
12. Driver receives reminders and notifications in mobile app

That is the complete MVP operational cycle.